

MovementLogger — Sensor & Module Catalog

9 Teile · 2026-05-21 08:09 UTC · BOM aus movement_logger_firmware · via crawl2pump

9 neu (7 Tage)

0 aktualisiert

Dev Boards 4

Raspberry Pi Zero 2 W (BCM2710A1, WiFi + BT 4.2 BLE)

OSS firmware

40-pin GPIO header

☐ USB-C

☒ WiFi

☒ Bluetooth

☐ GPS

☐ Motion sensors

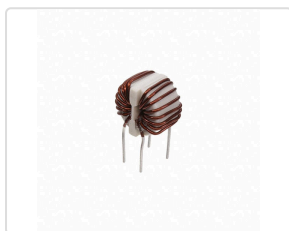
☒ SD-card

L×B×H 6.5 × 3 × 0.5 cm · 11 g

MCU: BCM2710A1 · Cortex-A53 quad-core @1 GHz · 512 MB RAM · Linux-host

Datasheet: <https://datasheets.raspberrypi.com/rpizero2/raspberry-pi-zero-2-w-product-brief.pdf>

OSS firmware: <https://github.com/raspberrypi/linux>



neu Raspberry Pi Zero 2 W (BCM2710A1, WiFi + BT 4.2 BLE) · SC0710 @ Farnell

CHF 2.89

Raspberry Pi Ltd @ Farnell · out of stock

Pi-Zero-footprint Linux SBC (65 × 30 × 5 mm). Quad-core Cortex-A53 @1 GHz, 512 MB RAM, microSD boot, 2.4 GHz WiFi + BLE 4.2, Mini-HDMI, Micro-USB power, 40-pin GPIO header. Hosts the PiSugar 3 UPS HAT (below) and the Waveshare L76X GPS HAT...

<https://ch.farnell.com/3290617>

GPS / GNSS 5

GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 4 × 4 × 1.3 cm · 80 g

Datasheet: <https://www.sparkfun.com/products/14986>



neu GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA) · GPS-14986 @ DigiKey

CHF 12.85

SparkFun @ DigiKey · out of stock

Active GPS antenna (~28 dB LNA, 3 m RG-174 coax, SMA male). Magnetic base — sits on the board's non-metal top or the case lid for best sky view. Needs a U.FL → SMA pigtail to mate with the MAX-M10S breakout's U.FL connector — see next...

<https://www.digikey.com/en/products/detail/sparkfun-electronics/14986/9682235>

Interface Cable SMA → U.FL, 100 mm (pigtail)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 10 × 0.3 × 0.3 cm · 3 g

Datasheet: <https://www.sparkfun.com/products/9145>



neu Interface Cable SMA → U.FL, 100 mm (pigtail) · SparkFun 9145 @ sparkfun.com

USD 5.75

SparkFun

100 mm pigtail: U.FL female (plugs into the SparkFun MAX-M10S breakout) ↔ SMA female bulkhead (mates with the antenna above's SMA male). Buy with the magnetic antenna or skip both if the onboard chip antenna is enough. · SMA / U.FL ·...

<https://www.sparkfun.com/products/9145>

Waveshare L76X Multi-GNSS HAT for Pi (Quectel L76B, UART, U.FL)

OSS firmware

40-pin GPIO header

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

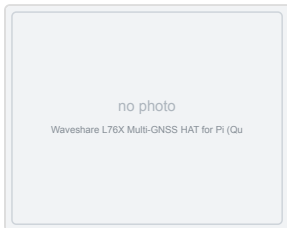
☐ Motion sensors

☐ SD-card

L×B×H 6.5 × 3 × 1 cm · 12 g

Datasheet: https://www.waveshare.com/wiki/L76X_GPS_HAT

OSS firmware: <https://github.com/raspberrypi/linux>



neu Waveshare L76X Multi-GNSS HAT for Pi (Quectel L76B, UART, U.FL) · L76X-GPS-HAT @ Waveshare (vendor page (reference))

Waveshare (manufacturer store)

Pi-HAT carrying a Quectel L76B GNSS module (GPS + BeiDou + QZSS, 1.575 GHz L1). Talks UART on the Pi's GPIO14/15 → /dev/serial0 → gpsd; ~30 mA, 1–10 Hz update rate. Patch antenna onboard plus a U.FL socket for an external active antenna...

<https://www.waveshare.com/l76x-gps-hat.htm>

Battery / Power 1

PiSugar 3 — 5000 mAh UPS HAT für Pi Zero (USB-C charge, no-solder)

OSS firmware

40-pin GPIO header

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 6.5 × 3 × 1 cm · 60 g

MCU: STM8S003F3 · STM8 8-bit @16 MHz · 8 KB flash · power-mgmt

Datasheet: <https://github.com/PiSugar/PiSugar/wiki/PiSugar3>

OSS firmware: <https://github.com/PiSugar/PiSugar>



neu PiSugar 3 — 5000 mAh UPS HAT für Pi Zero (USB-C charge, no-solder) · PiSugar @ PiSugar (vendor page)

PiSugar (manufacturer store) · in stock

Snap-on UPS for Pi Zero 2 W — zero soldering, pogo pins clip onto the GPIO underside. 5000 mAh LiPo onboard, USB-C charge input, hardware RTC (DS3231 + coin-cell backup), I²C heartbeat to the `pisugar- server` daemon for clean low-battery...

<https://www.pisugar.com/products/pisugar-3>

USD 39.99

Enclosure 9

Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 12 × 9 × 6 cm · 154 g

Datasheet: <https://www.hammmfg.com/electronics/small-case/plastic/1554.pdf>



neu Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm) · 1554G2GYCL @ DigiKey

Hammond Manufacturing @ DigiKey · in stock

Polycarbonate IP66 project box, external 120 × 90 × 60 mm, clear PC lid. Lid sealed with O-ring gasket and four corner screws. RF-transparent so onboard GPS (and any radar) reads through the wall. The Hall sensor reads a passing magnet...

<https://www.digikey.com/en/products/detail/hammond-manufacturing/1554G2GYCL/2359944>

CHF 22.77

SERPAC RBF63P06C10C — IP67 PC enclosure, clear lid (160 × 80 × 40 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

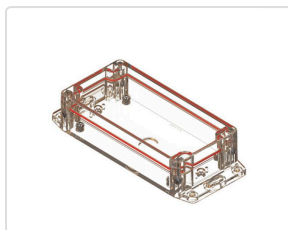
☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 16 × 8 × 4 cm · 105 g



neu SERPAC RBF63P06C10C — IP67 PC enclosure, clear lid (160 × 80 × 40 mm) · RBF63P06C10C @ DigiKey

CHF 26.97

SERPAC @ DigiKey · in stock

Same IP67 polycarbonate enclosure as the C16/C22 below in the shallow 40 mm variant (1.59 in depth) — the slimmest profile on the foilboard. ****Tight**** for the standard Pi-Zero + PiSugar + GPS-HAT stack: ~30 mm interior depth vs ~30–35 mm...

<https://www.digikey.com/en/products/detail/serpac/RBF63P06C10C/17100299>

SERPAC RBF63P06C16C — IP67 PC enclosure, clear lid (160 × 80 × 55 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 16 × 8 × 5.5 cm · 130 g



neu SERPAC RBF63P06C16C — IP67 PC enclosure, clear lid (160 × 80 × 55 mm) · RBF63P06C16C @ DigiKey

CHF 26.27

SERPAC @ DigiKey · in stock

Polycarbonate IP67 project box, external 160 × 80 × 55 mm (6.30 × 3.15 × 2.17 in), clear PC lid sealed with perimeter O-ring + stainless-steel screws into metal inserts. IP67 = rated for submersion to 1 m for 30 min — the Hammond is only...

<https://www.digikey.com/en/products/detail/serpac/RBF63P06C16C/22461984>

SERPAC RBF63P06C22C — IP67 PC enclosure, clear lid (160 × 80 × 85 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

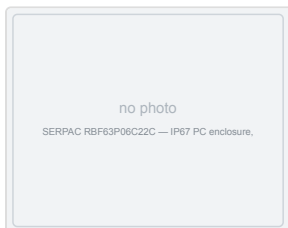
☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 16 × 8 × 8.5 cm · 165 g



neu SERPAC RBF63P06C22C — IP67 PC enclosure, clear lid (160 × 80 × 85 mm) · RBF63P06C22C @ SERPAC (vendor page (reference))

SERPAC (manufacturer store)

Same IP67 polycarbonate enclosure as the C16 above but in the deeper 85 mm variant (3.35 in depth) — gives ~45 mm headroom above the Pi-Zero stack, enough for a thicker LiPo (10 000 mAh PiSugar Pro, 18650 holder, etc.) or an extra HAT...

<https://www.serpac.com/product-by-series/rbf-series.html>